

Database Schema & SQL Guide

MySQL tables, MongoDB collections, queries, and sample data

1. Database Architecture Overview

K Star uses a polyglot persistence strategy. MySQL stores relational, ACID-compliant data (users, wallets, transactions, NACHA payments). MongoDB stores flexible document data that does not have a rigid schema (exchange rates, NFT metadata, conversion history).

Database	Type	Services	Data Stored
kstar (MySQL)	Relational	user, transfer, nacha	Users, wallets, transactions, ACH payments
kstar_exchange (MongoDB)	Document	exchange	Exchange rates, conversion history
kstar_nft (MongoDB)	Document	nft	NFT documents, trade history

2. MySQL Schema — kstar Database

■ users table

```
CREATE TABLE users (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  username VARCHAR(50) NOT NULL UNIQUE,  
  email VARCHAR(100) NOT NULL UNIQUE,  
  password VARCHAR(255) NOT NULL, -- BCrypt hash  
  phone VARCHAR(15) NOT NULL UNIQUE,  
  role ENUM('USER','ADMIN') DEFAULT 'USER',  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  INDEX idx_username (username),  
  INDEX idx_email (email)  
);
```

■ wallets table

```
CREATE TABLE wallets (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  user_id BIGINT NOT NULL,
```

```

currency VARCHAR(10) NOT NULL, -- INR, USD, EUR, GBP, JPY
balance DECIMAL(18,4) DEFAULT 0.0000,
upi_id VARCHAR(50) UNIQUE, -- phone@kstar (INR only)
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
UNIQUE KEY uq_user_currency (user_id, currency),
INDEX idx_upi_id (upi_id)
);

```

■ transactions table (transfer-service)

```

CREATE TABLE transactions (
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
  transaction_id VARCHAR(36) NOT NULL UNIQUE, -- UUID
  type ENUM('UPI_TRANSFER', 'WALLET_CREDIT', 'WALLET_DEBIT') NOT NULL,
  from_upi_id VARCHAR(50),
  to_upi_id VARCHAR(50),
  amount DECIMAL(18,2) NOT NULL,
  currency VARCHAR(10) DEFAULT 'INR',
  status ENUM('PENDING', 'PROCESSING', 'COMPLETED', 'FAILED') DEFAULT 'PENDING',
  description VARCHAR(255),
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  processed_at DATETIME,
  INDEX idx_from_upi (from_upi_id),
  INDEX idx_to_upi (to_upi_id),
  INDEX idx_status (status)
);

```

■ nacha_payments table

```

CREATE TABLE nacha_payments (
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
  payment_id VARCHAR(50) NOT NULL UNIQUE,
  type ENUM('CREDIT', 'DEBIT') NOT NULL,
  sec_code VARCHAR(10) NOT NULL,
  originator_name VARCHAR(100) NOT NULL,
  originator_routing_number VARCHAR(9) NOT NULL,
  originator_account_number VARCHAR(17) NOT NULL,
  receiver_name VARCHAR(100) NOT NULL,
  receiver_routing_number VARCHAR(9) NOT NULL,
  receiver_account_number VARCHAR(17) NOT NULL,
  account_type VARCHAR(10) NOT NULL,
  amount DECIMAL(18,2) NOT NULL,

```

```

currency VARCHAR(10) DEFAULT 'USD',
description VARCHAR(255),
status ENUM('PENDING', 'SUBMITTED', 'PROCESSING',
'SETTLED', 'RETURNED', 'FAILED') DEFAULT 'PENDING',
trace_number VARCHAR(15),
batch_number VARCHAR(20),
failure_reason VARCHAR(255),
initiated_by VARCHAR(50),
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
settled_at DATETIME,
effective_date DATETIME,
INDEX idx_payment_id (payment_id),
INDEX idx_initiated_by (initiated_by),
INDEX idx_status (status)
);

```

3. Useful SQL Queries

■ User Queries

```

-- List all users with wallet count
SELECT u.id, u.username, u.email, u.role, COUNT(w.id) AS wallet_count
FROM users u LEFT JOIN wallets w ON u.id = w.user_id
GROUP BY u.id;

-- Find user by UPI ID
SELECT u.*, w.currency, w.balance, w.upi_id
FROM users u JOIN wallets w ON u.id = w.user_id
WHERE w.upi_id = '9876543210@kstar';

-- Get total portfolio value per user (in original currencies)
SELECT u.username, w.currency, w.balance
FROM users u JOIN wallets w ON u.id = w.user_id
ORDER BY u.username, w.currency;

```

■ Transaction Queries

```

-- Get recent transactions for a UPI ID
SELECT * FROM transactions
WHERE from_upi_id = '9876543210@kstar'
OR to_upi_id = '9876543210@kstar'
ORDER BY created_at DESC LIMIT 20;

-- Daily transaction volume
SELECT DATE(created_at) AS date,
COUNT(*) AS count,

```

```

SUM(amount) AS total_amount
FROM transactions
WHERE status = 'COMPLETED'
GROUP BY DATE(created_at)
ORDER BY date DESC;

-- Failed transactions today
SELECT * FROM transactions
WHERE status = 'FAILED'
AND DATE(created_at) = CURDATE();

```

■ NACHA Queries

```

-- All pending NACHA payments
SELECT payment_id, type, sec_code, receiver_name, amount, created_at
FROM nacha_payments WHERE status = 'PENDING'
ORDER BY created_at;

-- Settlement summary by date
SELECT DATE(effective_date) AS settle_date,
sec_code, COUNT(*) AS count, SUM(amount) AS total_usd
FROM nacha_payments WHERE status = 'SETTLED'
GROUP BY settle_date, sec_code;

-- Return rate analysis
SELECT failure_reason, COUNT(*) AS count
FROM nacha_payments WHERE status = 'RETURNED'
GROUP BY failure_reason ORDER BY count DESC;

```

4. MongoDB Collections

■ ExchangeRate document (kstar_exchange.exchange_rates)

```

{
  "_id": "ObjectId(...)",
  "baseCurrency": "USD",
  "targetCurrency": "INR",
  "rate": 84.12,
  "source": "SCHEDULED_UPDATE",
  "updatedAt": "2024-11-15T10:30:00"
}

```

■ NFT document (kstar_nft.nfts)

```

{
  "_id": "ObjectId(...)",
  "tokenId": "NFT-20241101-0000001",

```

```
"name": "Golden Star #1",
"description": "Genesis K-Star NFT",
"imageUrl": "https://example.com/nft1.png",
"ownerUsername": "alice",
"creatorUsername": "alice",
"price": 0.5,
"currency": "ETH",
"status": "FOR_SALE", // MINTED | FOR_SALE | SOLD | BURNED
"mintedAt": "2024-11-01T12:00:00",
"tradeHistory": [
{ "from": "system", "to": "alice", "price": 0.5, "date": "2024-11-01" }
]
}
```